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Tobacco consumption, gender and caregiving: a case study in Argentina

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ABSTRACT

Introduction This study investigates the gender-specific economic impacts of tobacco price increases, focusing on their effects on caregiving time, female labour income and total family income within Argentine households.

Methods Using Argentine data, the research employs simulations to assess the potential changes in economic indicators following a hypothetical 25% increase in tobacco prices. The analysis considers specific behavioral assumptions to model household responses.

Results Simulations indicate that a 25% increase in tobacco prices is associated with a potential increase in the average monthly labour income of female caregivers by up to 8.6% in the preferred scenario. This effect is attributed to an increase in the hours worked by female caregivers and a rise in female labour force participation, driven by a reduced need for caregiving hours due to improved health outcomes or changes in household dynamics related to tobacco cessation. Consequently, this also implies a 0.53% increase in the monthly per capita household income (PCHI). These positive economic effects are non-linear across the income distribution, with women in the lowest quintile experiencing a more substantial 0.64% increase in PCHI.

Conclusions Tobacco price increases in Argentina may yield significant gender-specific economic benefits, particularly for female caregivers and low-income households. These findings highlight an additional economic rationale for implementing robust tobacco control policies, beyond direct health benefits, by demonstrating their potential to enhance female economic empowerment and reduce household poverty.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Smoking is associated with direct medical costs and productivity losses, which can have a substantial financial impact. It is commonly acknowledged that raising tobacco taxes is an effective way to lower tobacco use, promote cessation, prevent or delay initiation and reduce adverse effects on health and the economy.

WHAT THIS STUDY ADDS

⇒ This study fills the empirical evidence gap by analysing the effects of higher tobacco taxes on female labour supply. It builds on earlier work by examining how rising tobacco prices affect female labour participation, female income and per capita household income. Considering general equilibrium effects, tobacco taxes increase family income by reducing the time spent on caregiving, which women mainly supply. These effects are quantified using different data sources from Argentina.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The results provide new evidence on the socioeconomic effects of tobacco taxes, especially their consequences on female labour participation and household income. These findings underscore the role of an increase in tobacco taxes in gender parity and can inform policymakers about the potential advantages of higher tobacco taxes beyond health benefits. The study's methodology can also be applied to evaluate similar effects in other developing countries, thereby leading to a more comprehensive understanding of the impact of tobacco taxation.

INTRODUCTION

Tobacco use imposes a considerable economic burden on society regarding direct medical care and productivity losses.[1 2] A significant strand of the tobacco control literature has been dedicated to studying how the impact of a tobacco tax increase reduces its consumption, decreases the probability of smoking onset and increases cessation.[3–6] At the same time, part of the economic literature has focused on the relationship between caregiving tasks and labour supply and wages, considering that caregiving tasks have a high opportunity cost for caregivers.[7–15] However, a gap in the literature still needs to be filled on the links that increasing tobacco taxes may have on female labour participation and wages. In theory, a drop in the number of people with tobacco-related illnesses would result in fewer needs for in-house caregiving time and, therefore, in more time available to work. This study contributes to filling this gap. The methodological approach extends Fuchs and Meneses' approach by analysing the different channels through which the increase in tobacco taxes/prices may affect female and household incomes (ie, consumption, medical expenditures, labour supply).[16] Two novel channels are considered. First, the impact of higher tobacco taxes/prices is estimated on the extensive margin of female labour supply (ie, the number of women willing to work) and the intensive margin (ie, the average number of hours women work). Second, the effect of the extra labour supply on households' incomes is also estimated.

These channels are quantified using Argentinian data. Argentina still has a relatively high tobacco

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prevalence (22.2% in 2018; 26.1% for males and 18.6% for females).[2] As a consequence, tobacco is directly responsible for about 48 000 annual deaths, and the annual cost of treating tobacco-related illnesses is equivalent to 0.6% of the gross domestic product.[17] Despite its level of per capita income, Argentina has a relatively low female participation rate compared with other Latin American countries. For 2019, the female participation rate in Argentina was 49.4% of the working-age population, while the average in Latin America (including Argentina) was 51.8%.[18] These facts (high tobacco prevalence and low female labour participation) make Argentina an attractive case study.

DATA AND METHODS

Several nationally representative surveys are used to estimate the models. The main one is the National Use of Time Survey of 2021 (ENUT for its acronyms in Spanish). It provides information on the intrahousehold distribution and the number of hours devoted to caregiving responsibilities. ENUT reports the number of hours dedicated to the caregiving of specific age groups rather than the number of hours devoted to caregiving for the sick. Two additional surveys are used to estimate the impact of tobacco-related illnesses on caregiving time, the 2018 National Risk Survey (ENFR for its acronyms in Spanish) and the 2018 National Study on Profiles of Disabled People (ENPPD for its acronyms in Spanish), which provide information on the prevalence of people who smoke across different sociodemographic groups. Finally, the 2021 Permanent Household Survey (EPH for its acronyms in Spanish) is used to estimate female income and per capita household income (PCHI). PCHI was estimated using LASSO (least absolute shrinkage and selection operator), since income data are not available in the ENUT database to the public (see online supplemental table A1). LASSO introduces bias by design, as it shrinks some coefficients towards zero in exchange for reducing variance. This trade-off is appropriate in the context of this study, where the goal is prediction, and where lower variance improves the overall reliability of the imputed values.[19–21]

The methodology for estimating the effects of tobacco price increases on caregiving, particularly regarding tobacco-related diseases, is based on previous research linking the impact of tobacco prices on tobacco prevalence, medical expenses and working life years.[16] The equation that links increases of tobacco prices to female labour monthly income (Y_f) is:

$$\Delta Y_f = \left(\left(\frac{\Delta P}{P} \times \varepsilon \times \mathrm{ISP} \right) \times \mu \times L_{female} \times \bar{W}_f \times \varphi_{penalisation} \right) + \left(\left(\frac{\Delta P}{P} \times \varepsilon \times \mathrm{ISP} \right) \times \gamma \times \varepsilon_{care} \times C_T \times \eta \times L_{care} \times \bar{W}_f \right) \quad (1)$$

where $\frac{\Delta P}{P}$ is the percentage tobacco price variation; ε represents the demand own-price elasticity of the tobacco prevalence (ie, the percentage change in the number of people who smoke when tobacco price increases 10%); ISP denotes the change in the tobacco-related ill individuals relative to a change in the number of people who smoke; μ is the change in the number of employed women relative to the change in the total number of ill individuals; L_{female} is the number of female workers; and $\bar{W}_f$ is the average female monthly wage. The monthly wage is adjusted by a penalty factor to reflect either that women re-entering the labour market after providing unpaid care may earn less due to skill depreciation accumulated during their time out of the workforce, or that they may still face time constraints from ongoing caregiving responsibilities, which lead them to seek more flexible but lower paid jobs.[22]

In the second term, γ captures the elasticity of caregiving hours related to tobacco with respect to the total number of care needers in a household (ie, the percentage increase/decrease in caregiving hours when the number of people needing care in a household increases by 10%); ε_{care} represents the elasticity of substitution between hours allocated to caregiving and hours allocated to work; C_T is the aggregate number of hours devoted to tobacco-related caregiving; η is the semielasticity of female working hours to female labour income (ie, the percentage increase in female labour income when working hours increase by a unit); and finally, L_{care} is the pre-existing number of female caregivers.

The underlying assumption in equation 1 is that a proportion of the time not allocated to caregiving is assigned entirely to external paid work. This extra time can be separated into two parts: first, it increases female labour participation of women who were previously not working (ie, extensive margin); second, it increases the number of hours worked among those who participate (ie, intensive margin). The extensive margin is estimated in the first term of equation 1 that shows how increasing tobacco prices decreases the number of tobacco-related care needers ($\frac{\Delta P}{P} \times \varepsilon \times \mathrm{ISP}$), and how, in turn, this affects the number of employed women (through μ, the elasticity of care needers to the number of employed women). It is assumed that female caregivers entering the labour market earn, on average, 28% less than non-caregiver women, based on recent evidence from Chile, indicating that women experience this wage penalty on having a child, which it is assumed to be similar for other caregiving activities.[22]

Women are assumed to get the average female monthly wage ($\bar{W}_f$). The second term estimates the intensive margin by multiplying the impact of higher tobacco prices on the average hours of female caregiving ($\frac{\Delta P}{P} \times \varepsilon \times ISP \times \gamma \times \varepsilon_{care} \times C_T$) (where it can be assumed an imperfect substitution of caregiving hours into labour hours) and how higher working hours impact female labour income (through η, the semielasticity of hours worked to labour income). Then, the aggregate effect is computed by multiplying this effect by the total number of female caregivers and by the average wage of female workers.

The elasticity ε in equation 1, which is half the total own-price elasticity, following global estimates,[23] is taken from several studies. Three alternative scenarios for total own-price elasticities are considered (see online supplemental table A2): (1) a 'lower-bound' elasticity, with an average value of −0.262[24]; (2) a 'medium-bound' elasticity, with an average value of −0.38[24]; and (3) an 'upper-bound' elasticity, with an average value of −0.625.[2 24] The latter, which serves as the preferred specification, is based on Argentine household expenditure data and is not transferred from other country contexts, which ensures that the elasticity estimate reflects local behavioural and market conditions.[2]

The other variables required to estimate equation 1 come from different sources. $\bar{W}_f$ is taken from the 2021 EPH; the parameter ε_{care} is estimated for Brazil[25]; C_T is taken from the 2021 ENUT; the parameter ISP is taken from Pichon-Riviere *et al*[17]; $\varphi_{penalization}$ is set to 0.72 (although in online supplemental table A6 the results of the model removing the wage penalty are also included); and η, μ and γ are estimated from the following regressions:

$$\ln(\mathrm{LY}_i) = \alpha + \eta\,\mathrm{HW}_i + \beta X_i + \epsilon_i \quad (2)$$

$$FW_i = \alpha + \mu\, ln\,(CNI_{ic}) + \beta X_i + \epsilon_i \quad (3)$$

$$\ln(\mathrm{CH}_i) = \alpha + \gamma \ln(\mathrm{CNI}_{ic}) + \beta X_i + \epsilon_i \quad (4)$$

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where LY_i represents the labour monthly income of individual i; HW_i represents the number of hours worked per month by individual i; FW_i is a dichotomous variable which indicates whether female i is working or not; CNI_i represents the number of care needer individuals in the household (taking into account that the variable CNI_i is corrected by an estimate of individuals who have tobacco-related illnesses, using the 2018 ENFR and the 2018 ENPPD); CH_i is the number of hours per month that individual i dedicates to caregiving; X_i is a vector of control variables; and ϵ_i is the error term. The ENUT is used to estimate equations 3 and 4, while the EPH is used to estimate equation 2, since ENUT does not include information regarding income, or the total number of hours worked per month by the respondents. Results of these estimations can be found in online supplemental tables A3–A5. The parameter estimated for Brazil and used here considers the effect of a childcare reform that made preschool attendance mandatory.[25] Such a reform implied that women increased their working time by 26 hours/week. Assuming that preschool attendance covered 40 hours/week (8 hours/day), this would imply a caregiving-to-labour elasticity of 0.65. While this estimate refers to reductions in childcare time, it is assumed that a similar elasticity applies to caregiving for ill individuals, under the notion that both types of unpaid care act as substitutes for formal labour supply.

The simulated increase in tobacco prices due to an increase in tobacco taxes ($\Delta P/P$) is set to 25%. This change is not arbitrary since it is similar to the change in the price of cigarettes that Argentina experienced after its recent changes in tobacco taxes.[26] This change is also similar to that simulated in other studies for Argentina[2–8 27 28] and other Latin American countries.[6–14 16]

To address the uncertainty introduced by combining several estimated parameters in a multiplicative structure, a full non-parametric bootstrap procedure on equation 1 was implemented. Specifically, 999 bootstrapped samples from the underlying microdata are drawn, and the relevant regressions used to obtain parameters η (equation 2), μ (equation 3) and γ (equation 4) in each iteration are re-estimated. These parameters are then used to compute the simulated impacts via equation 1. This approach captures the sampling variability of each parameter as well as the compounded uncertainty from their interaction in the simulation. The resulting distributions of simulated outcomes across all iterations are used to construct empirical CIs, which are reported in the tables (eg, table 3). This procedure also allows for correlations across parameter estimates (eg, between η, μ and γ), thus providing a more robust account of the total uncertainty in the simulated impacts.

Finally, to measure the effect of the tobacco price increase on households' incomes due to the increase in female labour incomes, the following equation is estimated:

$$\frac{\Delta \text{PCHI}}{PCHI} = \left(\frac{\Delta P}{P} \times \varepsilon \times \text{ISP}\right) \times (\gamma \times \theta) \quad (5)$$

where $\frac{\Delta PCHI}{PCHI}$ is the percentage increase in the per capita monthly household income; and θ represents the elasticity of the PCHI with respect to caregiving hours per day (ie, the percentage change in the PCHI when caregiving by a member of the household increases by 10%). As in equation 1, the first term estimates the reduction in tobacco-related care needers due to an increase in tobacco prices, while the second term measures how caregiving changes to care needer reduction, and then how PCHI relates to caregiving. Overall, equation 5 links the percentage increase in tobacco prices (due, for instance, to a tobacco tax increase) with the percentage increase in PCHI through the effects on caregiving time.

Equation 5 requires the additional estimation of θ, which is estimated through the following model:

$$\ln\left(\text{PCHI}_{\text{ic}}\right) = \alpha + \theta \ln\left(\text{CH}_i\right) + \beta X_i + \epsilon_i \quad (6)$$

where CH_i is the number of hours per month that individual i dedicates to caregiving; $PCHI_{ic}$ is the PCHI of individual i living in family c; X_i is a vector of control variables; and ϵ_i is the error term. Since θ is in log-log format, the percentage variation was obtained. The 2021 ENUT is used to estimate this equation (results are shown in online supplemental table A5). Equations 4 and 6 were estimated using seemingly unrelated regressions, given the potential correlation of their residuals. On the other hand, previous literature on tobacco demand in Argentina includes estimates of own-price elasticities by income quintiles.[2 24 29] These estimates can disentangle the distributional effect of tobacco price changes on female labour incomes and the PCHI.

To compute the CIs for the simulated effects of this case, the delta method, a statistical technique used to approximate the variance of a non-linear function of estimated parameters, was applied. In this case, since the simulation combines γ and θ multiplicatively, the delta method allowed to derive the SE of their product using a first-order Taylor expansion around the estimated values. This provides a more accurate measure of the uncertainty in the final simulated effect, accounting for the variability in both parameters.

It is important to clarify that these estimations represent long-run equilibrium outcomes—that is, the comparison of equilibriums in two steady-state scenarios before and after a sustained reduction in tobacco-related illnesses caused by a price

Table 1 Proportion of caregivers by predicted PCHI quintiles and gender

Quintiles of predicted PCHI	Total	Men	Women
Q1	6.4% (total: 282 075)	3.8% (total: 72 115)	8.4% (total: 209 960)
	(SD: 24.4%)	(SD: 19.0%)	(SD: 27.7%)
Q2	7.9% (total: 350 071)	8.2% (total: 183 552)	7.7% (total: 166 519)
	(SD: 27.0%)	(SD: 27.4%)	(SD: 26.6%)
Q3	5.8% (total: 255 210)	5.6% (total: 124 458)	5.9% (total: 130 752)
	(SD: 23.3%)	(SD: 23.1%)	(SD: 23.6%)
Q4	5.3% (total: 233 072)	4.8% (total: 104 933)	5.8% (total: 128 139)
	(SD: 22.4%)	(SD: 21.3%)	(SD: 23.3%)
Q5	3.8% (total: 168 121)	2.7% (total: 56 310)	4.7% (total: 111 811)
	(SD: 19.3%)	(SD: 16.3%)	(SD: 21.3%)

Distribution of caregivers by gender and by quintile of household per capita income. Own elaboration using data from ENUT and EPH.
ENUT, National Use of Time Survey of 2021; EPH, 2021 Permanent Household Survey; PCHI, per capita household income.

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Table 2 Descriptive statistics by gender

A. Caring hours by gender				
			Women caregivers	Men caregivers
Number of women/men who dedicate at least 1 hour to caring			774 168	566 007
Women/men who dedicate at least 1 hour to caring with respect to population			6.4% (SD: 24.5%)	5.1% (SD: 22.0%)
Monthly hours dedicated on average to caring for tobacco-related patients			32.0 (SD: 31.8)	16.3 (SD: 16.1)
B. Caring hours by gender and education level				
Education level	**Caring women (weighted)**	**Average caring hours per month**	**Caring men (weighted)**	**Average caring hours per month**
Low skilled	278 213	35.7 (SD: 37.1)	174 144	18.9 (SD: 18.6)
Medium skilled	204 249	33.4 (SD: 28.3)	237 654	15.1 (SD: 15.9)
High skilled	290 709	27.6 (SD: 28.3)	154 209	15.0 (SD: 12.9)
Total	774 168	32.0 (SD: 31.8)	566 007	16.3 (SD: 16.1)

Own elaboration using data from ENUT. Results are weighted according to the sample weight. Exposed time is measured in hours, where caregiving hours were adjusted based on the proportion of individuals who smoke according to ENFR and ENPPD, to reflect the hours dedicated to caring for tobacco-related patients. Low skilled: incomplete secondary education or less. Medium skilled: complete secondary education. High skilled: incomplete tertiary education or higher. SDs in parentheses.
ENFR, 2018 National Risk Survey; ENPPD, 2018 National Study on Profiles of Disabled People; ENUT, National Use of Time Survey of 2021.

increase. The transitional dynamics or the specific timing of health improvements, which may involve significant lags due to the long development period of smoking-related diseases, was not modelled. Instead, the focus is on quantifying the long-run effects on caregiving, female labour supply and household income, once those health effects are fully realised.

Despite the inclusion of extensive control variables in the estimation of equations 2–6, it is important to acknowledge the likely persistence of endogeneity concerns. These include simultaneity, omitted variable bias and reverse causality, which may affect the interpretation of the estimated coefficients. For example, caregiving responsibilities and labour force participation may be jointly determined by unobserved factors, and caregiving intensity may influence, and be influenced by, income constraints. Considering these limitations, each estimated parameter should be interpreted as a conditional correlation rather than a causal effect. The simulation exercise should therefore be seen as an exploration of potential magnitudes under a set of empirically grounded behavioural associations, rather than as a projection of definitive causal impacts. Given the use of cross-sectional data, stronger identification strategies—such as those relying on fixed effects—are not feasible within this empirical framework.

RESULTS

Table 1 displays the distribution of caregiving time by the predicted income quintiles (Q1 is the quintile with the lowest PCHI). It shows that in almost all quintiles (except in quintile 2), the share of female caregivers is higher than that of males. It also indicates that caregiving is monotonically decreasing across quintiles, with Q1 being almost 70% higher than in quintile 5.

Table 2A indicates that approximately 6.4% of women participate in caregiving activities, compared with 5.1% of men. The average monthly hours devoted by women to tobacco-related caregiving (adjusting the care hours of the ENUT to the share of people who smoke in Argentina obtained from other sources) double those devoted by men (32 hours/month for women vs 16.3 hours/month for men). Table 2B shows that low-skilled women stand out as dedicating significantly more time to caregiving than women with medium and high levels of skill.

The effects of the tobacco price increase (25%) on female labour income (as shown in equation 1) are displayed in table 3. As said before, it is assumed that women dedicate their previously allocated time to tobacco-related caregiving to work. Column 1 of table 3 presents three scenarios of increases in female labour

Table 3 Main effects on female working hours and total female monthly income using a 25% price increase of tobacco

Elasticity bound	Number of new female workers (absolute variation)	Mean number of monthly hours worked increase (absolute variation)	Total female labour monthly income increase by extensive margin (in %, for caregivers)	Female labour income increase by extra hours worked (in %, for caregivers)	Percentage variation of total female labour monthly income for women caregivers
	(1)	(2)	(3)	(4)	(5)
Lower	13 140 (1759 to 26 079)	0.32 (0.20 to 0.44)	3.17% (0.21% to 6.29%)	0.44% (0.26% to 0.59%)	3.61% (0.64% to 6.70%)
Medium	19 058 (2551 to 37 825)	0.47 (0.29 to 0.63)	4.60% (0.30% to 9.12%)	0.63% (0.38% to 0.85%)	5.23% (0.92% to 9.71%)
Upper (preferred)	31 346 (4196 to 62 212)	0.77 (0.48 to 1.04)	7.56% (0.50% to 15.01%)	1.04% (0.63% to 1.40%)	8.60% (1.52% to 15.97%)
Baseline	6 181 512 (total)	32 (mean)	US$135 328 287 (total)	US$135 328 287 (total)	US$135 328 287 (total)

Source: Own elaboration.
Column 3 reports the total labour income increase driven by the entry of new female workers into the labour market. Column 4 shows the income increase attributable to additional hours worked, while column 5 reflects the combined effect of columns 3 and 4. Columns 1 and 2 report the absolute increase in the number of female workers and in the number of monthly hours worked. The last row shows the baseline number for each of the variables of interest. Estimations are made by applying a 25% variation in tobacco prices. Estimates are made using ENUT, EPH and other alternative sources. The exchange rate is weighted equally between the official exchange rate and the informal exchange rate for the month of August 2021 (US$1=ARS141.34). The elasticities that allow us to carry out a differentiated analysis by quintile are those of online supplemental table A2. The 90% empirical CIs are displayed in parentheses, which were computed via 999 bootstrap iterations.
ARS, Argentine peso; ENUT, National Use of Time Survey of 2021; EPH, 2021 Permanent Household Survey.

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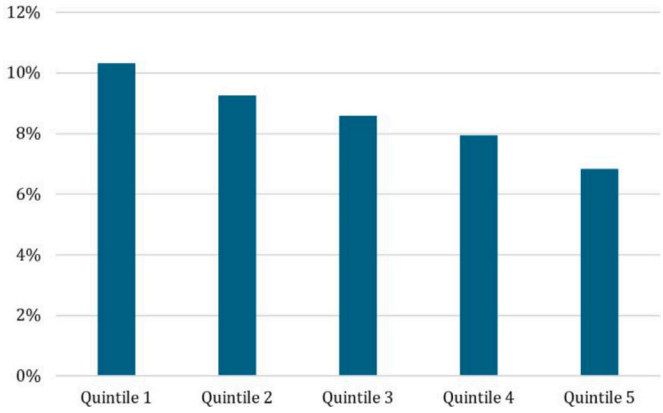

Figure 1 Female total labour income change by quintiles.

participation resulting from a decrease in tobacco-related caregiving (extensive margin), while column 2 illustrates an increase in the number of working hours (intensive margin). Column 3 indicates the increase in female labour income due to the increase in female labour participation of caregivers (extensive margin), which ranges from 3.17% to 7.56%, while column 4 shows that for the intensive margin. Overall, the simulated effects suggest that female labour income could increase by between 3.61% and 8.60%, depending on the scenario; however, these magnitudes should be interpreted with caution due to the underlying modelling assumptions. The latter should be interpreted as an upper bound impact under the model's assumptions, which is, however, the preferred specification because the elasticity used was estimated for Argentina. A decomposition of such an increase for the preferred scenario by quintiles is presented in figure 1. The results indicate a progressive effect, with a potential impact ranging from 10.3% for the lowest quintile to a 6.8% increase for the highest quintile.

Finally, table 4 shows the effect that a 25% increase in tobacco prices has on the PCHI by quintiles (equation 5). The increase in female labour incomes is associated with this effect. In all cases, there are increases in the PCHI (the average increase ranges from 0.2% to 0.5% of the PCHI), but they are higher for the lowest quintile (ranging from 0.2% to 0.6%).

DISCUSSION

Tobacco consumption has several negative social consequences. The most studied ones relate to individual health status, the social cost of illnesses and other issues.[30 31] Indirect consequences, though important in economic terms, are less studied. This study focuses on the consequences of tobacco consumption on female labour outcomes.

The main results show that rising tobacco prices in Argentina are associated with rising household incomes by increasing female incomes. A 25% tobacco price increase could raise female labour participation by up to 0.51% (extensive margin), while the total number of extra hours worked could increase by up to 2.4%. Overall, the monthly labour income of females involved in caregiving could increase by up to 8.6% under the model's assumptions. This average masks a progressive increase in female labour incomes, as the lowest quintile would experience an income increase that is up to 50% higher than that of the highest quintile. These results add a gender and distributional component to the already compelling justifications supporting policies to increase tobacco prices (eg, taxes).[6 31 32] The increase in female labour income is sensitive to the values of several parameters, including the elasticity of substitution between caregiving and working hours (ε_{care}) or the wage penalty for new female workers ($\varphi_{penalization}$). For instance, changing the value of ε_{care} from the 0.65 used in the models to 1 (perfect substitution between caregiving and working hours for women who were already in the labour force) and the value of $\varphi_{penalization}$ from the 0.72 to 1 would imply that the simulated tobacco price change increases female labour income by 12.1%.

Likewise, the effects of rising tobacco prices on the PCHI show that a 25% rise is associated with a 0.53% increase in the PCHI, considering only changes in female labour outcomes. This may appear minor at first glance, but it should be analysed in relative terms. The reduction in the number of tobacco-related sick individuals needing care is only 5.2%, so the effect of 0.53% in that context is particularly large.

The results do not account for some potentially important effects that should be considered, which could enhance the social benefits of increasing tobacco prices. First, the effects on male labour outcomes and their impact on PCHI should be considered. While caregiving mainly falls on women, men also provide a non-negligible share of care. According to ENUT data, approximately 5.1% of adult men report caregiving activities, devoting an average of 16.3 hours/month to tobacco-related care, about half the average time reported by female caregivers. Although the conducted simulations do not explicitly include these male caregivers, this information helps contextualise the potential magnitude of effects that may be omitted. Given that male workers in Argentina tend to earn higher wages on average, reallocations of time from caregiving to paid work among men could also translate into additional gains in household income.

Second, the results do not consider the effects on females' accumulated human capital. An increase in labour participation

Table 4 Effects of a 25% increase in tobacco prices on PCHI of decreasing caregiving time for female caregivers by quintiles

Elasticity bound	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5	Mean
Lower	0.27% (0.20% to 0.34%)	0.23% (0.17% to 0.29%)	0.22% (0.16% to 0.28%)	0.20% (0.15% to 0.26%)	0.19% (0.14% to 0.23%)	0.22% (0.17% to 0.28%)
Medium	0.52% (0.39% to 0.65%)	0.42% (0.31% to 0.53%)	0.32% (0.24% to 0.40%)	0.23% (0.17% to 0.28%)	0.13% (0.09% to 0.16%)	0.32% (0.24% to 0.40%)
Upper (preferred)	0.64% (0.47% to 0.80%)	0.57% (0.43% to 0.72%)	0.53% (0.40% to 0.67%)	0.49% (0.37% to 0.62%)	0.42% (0.31% to 0.53%)	0.53% (0.40% to 0.67%)
Baseline PCHI (mean)	$15 561	$27 522	$36 565	$48 316	$65 210	$34 949

Source: Own elaboration.
Proportional changes in per capita family income estimated for women, applying a 25% variation in tobacco prices. Estimates are made using ENUT, EPH and other alternative sources. The elasticities that allow us to carry out a differentiated analysis by quintile are those of online supplemental table A2. The last row shows the baseline number of PCHI in ARS for each quintile. The 95% CIs are displayed in parentheses.
ARS, Argentine peso; ENUT, National Use of Time Survey of 2021; EPH, 2021 Permanent Household Survey; PCHI, per capital household income.

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would imply, among other things, more on-the-job training, which would increase future productivity and income.

Third, they do not consider the value of intangibles, such as the increase in social welfare associated with women's empowerment. Increased female labour participation may also significantly affect household welfare and children's health outcomes.[33–35] Fourth, other intangibles, such as women's self-perceived social contribution, mental health status, domestic violence, etc, are not considered.

The study has some limitations. First, the estimated effects are static and based on observational associations. A price change is simulated, and the impact of this change is calculated for several variables (female participation, PCHI, etc) in the new equilibrium. There is no account of the potential effects occurring until such an equilibrium is reached. Second, many second-round or higher order effects related to women and households' well-being are not included. In some cases—such as long-term human capital accumulation or changes in self-perception—they are omitted due to modelling or data limitations. In other cases, they are likely to be quantitatively negligible. For instance, evidence from Argentina estimates the income elasticity of smoking prevalence at −0.007, suggesting that increases in female labour income have virtually no effect on tobacco consumption.[2] Third, due to data limitations, certain estimates rely on parameters estimated for other countries. For instance, the study applies an elasticity parameter from a study on childcare reform in Brazil to the context of caregiving for individuals with illnesses in Argentina. Institutional factors affecting labour markets could imply different results across countries regarding female labour supply. Further research should be conducted to enquire about this context-specific elasticity for this type of caregiving. Fourth, these are simulated impacts based on several different databases and equations. Statistical errors arising in the several estimated equations can be significant. However, the direction of the results is consistent with what economic theory predicts. Fifth, the simulation relies on parameters estimated through OLS (ordinary least square) regressions, which are subject to endogeneity concerns. Therefore, these estimates should not be interpreted as definitive causal effects, but rather as plausible magnitudes under specific behavioural assumptions.

CONCLUSION

The study provides evidence, based on simulations, that an increase in tobacco prices in Argentina could potentially benefit women by reducing the caregiving time for tobacco-related illnesses, thus allowing for greater female labour participation and reducing gender inequality. The quantitative impacts on female labour supply can be significant and can positively impact household per capita income, particularly for low-income households. These findings could inform policy discussions about tobacco control and its social and economic implications.

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